CDEQ-V8.0 Dual-Master-Equation Electromagnetic Pulse Propulsion System: Theory and Hardware Implementation

中文：CDEQ-V8.0 双主方程电磁脉冲推进系统：理论与硬件实现

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Abstract

This work presents the CDEQ-V8.0 dual-master-equation electromagnetic pulse propulsion system, a physically consistent and hardware-realizable framework for deep-space propulsion. The system adopts a dual-master-equation paradigm: the V7.0 master equation for functional optimal navigation and the V8.0 master equation for electromagnetic thrust generation.

The theoretical foundation is anchored by Einstein's mass-energy equivalence, Maxwell's electromagnetic equations, and Noether's theorems, ensuring rigorous physical validity. The immutable core modules—PCTT-CW topology, CENHE-LEX constraint unit, and GCVT-10 global convergence interlock—guarantee structural stability, deterministic operation, and immunity to divergence and pseudo-stability.

The system is fully synthesizable in Verilog HDL with fixed-point arithmetic and deterministic timing. It realizes inertial-sustained, pulse-superposed acceleration without energy recycling or propellant consumption, complying with aerospace safety standards. This work provides a canonical, extensible foundation for next-generation interstellar propulsion systems.

1 Problem Statement

Conventional aerospace propulsion systems rely on continuous thrust, propellant consumption, or energy-recycling mechanisms, leading to inherent limitations in efficiency, operational duration, and deep-space adaptability. Existing architectures lack a stable topological foundation, a provably convergent control mechanism, a physically consistent navigation paradigm, and a high-efficiency thrust generation scheme. This work addresses these critical deficiencies within a mathematically closed and physically self-consistent framework.

2 Theoretical Analysis

The system is grounded in five fundamental disciplines that constitute its theoretical invariance:

1. Einstein's Relativity: Defines the relativistic limit and velocity constraints.
2. Maxwell's Electromagnetism: Governs field evolution and momentum transport.
3. Noether's Theorem: Verifies symmetry and conservation law compliance.
4. Newtonian Mechanics: Establishes inertial persistence and impulse dynamics.
5. Lyapunov Stability: Provides rigorous criteria for global convergence and structural robustness.

All operational principles and topological designs are derived under these constraints, ensuring physical legitimacy and engineering realizability.

3 Core Foundations of CDEQ Series (Unchangeable Soul)

3.1 PCTT-CW Topology: Adjacent Reinforce, Interval Restrain

PCTT-CW (Penta-Cyclic Clockwise Topology) is the uniquely stable minimal cyclic structure. It enforces fixed local interaction rules:

- Adjacent nodes: mutual reinforcement for sustained state evolution
- Nodes separated by one interval: mutual restraint to prevent state overflow

Reinforcement ensures driving vitality; restraint eliminates unbounded divergence. Only a 5-node structure satisfies both conditions simultaneously.

3.2 CENHE-LEX: Rigid Constraint & Law Enforcement Unit

CENHE-LEX is the highest-priority core for structural stability and state conservation:

1. Fixes the central state to eliminate drift
2. Preserves global topological conservation
3. Restrains overloaded states and enforces recovery
4. Prohibits non-compliant or unstable operation

This unit guarantees the system never diverges, collapses, or runs uncontrolled.

3.3 GCVT-10: Global Convergence Verification & Safety Interlock

GCVT-10 is a mandatory 10-step consecutive stability verification mechanism.

Cost function:
$\Phi = \Sigma[\ ||D(t) - 0.8\cdot Dev(t)||^2 + 0.2\cdot||P(t) - P(t-1)||^2\ ]$

Convergence conditions (valid for 10 consecutive steps):

1. Central stability: $|E_{11} - 5.0| < \varepsilon$
2. Structural balance: $\max(|\text{row sum} - 15|, |\text{column sum} - 15|) < \varepsilon$
3. Cost threshold: $\Phi < \Gamma$

GCVT-10 eliminates all pseudo-stability and false activation.

4 Dual Master Equations: Theoretical Core

4.1 V7.0 Master Equation: Optimal Navigation Decision

$$J = \int_{t_0 \to t_f} [\alpha \cdot R(t) + \beta] dt + \gamma \cdot \|X(t_f) - X_{target}\|^2$$

where

- J: global cost functional
- α, β, γ: weighting coefficients
- R(t): path resource cost
- X(t_f): terminal state
- X_target: target state

This functional governs trajectory optimization, resource consumption, temporal cost, and terminal positioning precision. It serves as the decision-making brain of the system.

4.2 V8.0 Master Equation: Electromagnetic Thrust Execution

$$F = 2 \cdot \eta \cdot A \cdot E \cdot H / c$$

where

- F: electromagnetic impulse thrust
- η: dimensionless energy efficiency ($\eta = 0.81$)
- A: effective radiation area
- E: spatially averaged electric field strength
- H: spatially averaged magnetic field strength
- c: speed of light in vacuum

This equation describes the conversion of electromagnetic energy flow into directional mechanical impulse via Poynting momentum transport. It acts as the executive muscle that performs physical propulsion.

The dual master equations form a fully decoupled yet highly coordinated closed loop: V7.0 commands the jump; V8.0 executes the jump.

5 Engineering Implementation

The system is fully synthesizable in Verilog HDL, with a 167 ms global clock and Q14 fixed-point arithmetic. Key engineering characteristics include:

- Deterministic timing and zero floating-point operations
- No divergent feedback or unstable resonance
- No energy recycling or propellant consumption
- Inherent robustness under extreme environmental conditions
- Compliance with IEC 61508 SIL-4 and DO-254 aerospace standards

The V-shaped monocoque structure enhances rigidity, attitude stability, and electromagnetic symmetry.

6 De-Risking & Physical Consistency Verification

The system is rigorously verified to eliminate pseudo-science and invalid physical assumptions:

- No perpetual motion mechanisms
- No spontaneous energy creation
- No superluminal velocity assumptions
- No violation of conservation laws
- No unprovable topological or dynamic claims

All operational behaviors conform to relativity, electromagnetism, and Lyapunov stability criteria.

7 Conclusion

The dual-master-equation architecture establishes CDEQ-V8.0 as a physically consistent, hardware-realizable, and high-efficiency propulsion system. Supported by the immutable PCTT-CW, CENHE-LEX, and GCVT-10 core modules, the system achieves stable, inertial-sustained, pulse-superposed acceleration for deep-space navigation.

This work provides a universal, extensible foundation for interstellar mobility, defining a rigorous paradigm from theoretical formulation to engineering implementation.

8 Sovereignty Declaration

All theoretical formulations, topological structures, dual master equations, hardware architectures, naming conventions, and design methodologies of the CDEQ system are the original intellectual property of Jiang Wenjia. Any derivation, extension, or application must formally cite the CDEQ series.

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The Dao of CDEQ

Heart as Compass, Sword of Readiness.
Seek no trouble, fear no trouble.
Sword in sheath, hand on hilt.
Heart as anchor, mouth without noise.
Ignore us now.
The gate was never open.